



Course:	Object Oriented Programming (OOP)	Semester:	Spring 2026
Section:	BCS-2G	Total Marks:	100
Assignment - 2		Deadline:	22nd March, 2026

Submission Instructions

(Please read the following instructions carefully before submitting the assignment).

1. Save your file with your Roll Number (format: xxL-xxxx_A1.cpp).
2. It should be clear that your assignment will not get any credit if:
 - The assignment is submitted after the due date.
 - Assignment is plagiarized from any source. (online/other students/etc.)

Important Notes:

- i. There should not be any memory leakage in your program, otherwise marks will be deducted.
- ii. Your program must be as modular as possible. Your main() should be a sequence of function calls.
- iii. You are not allowed to use global variables and go-to instruction.
- iv. All allocations of pointers should be dynamic. Use of static arrays is prohibited.
- v. Remember to delete the allocated memory when it is no longer needed.
- vi. All the data will be given by the user. Do not submit a program that runs on hard coded inputs.
- vii. Pass the pointers to functions instead of subscript notation [].
- viii. Make proper functions to solve the problems.
- ix. Debug your code extensively to find errors/bugs. Test modules against the sample input given in the assignment.
- x. Use of "string" datatype and all built-in functions like strcmp() and strtok() etc. is strictly prohibited.
- xi. Write readable, well-formatted and well-commented code.
- xii. No matter how hard things seem, be confident and believe in your own abilities.
- xiii. Do not panic and remember to take breaks from your device screen while attempting this assignment.

Q1. Hogwarts SpellCraft System

At Hogwarts School of Witchcraft and Wizardry, students constantly learn and experiment with magical spells. Each spell has magical properties like power, mana cost, and difficulty. Students can learn new spells, combine spells to create stronger ones, and compare spells to determine which one is stronger. Students store their learned spells inside a magical SpellBook. Your task is to design a SpellCraft System using Object-Oriented Programming to manage spells and spellbooks.

Part 1 - Spell Class

Create a class called Spell with the following private attributes:

```
string spellName  
int power  
int manaCost  
int difficulty
```

Implement:

- Default constructor
- Parameterized constructor
- Getter functions if needed
- Create the function: void displaySpell()

Example Output:

Spell: Expelliarmus

Power: 40

Mana Cost: 20

Difficulty: 3

Part 2 - SpellBook Class

Each Hogwarts student owns a SpellBook that stores the spells they have learned.

Private attributes:

```
Spell* spells  
int capacity  
int totalSpells  
string ownerName
```

- Implement the constructor: SpellBook(string ownerName, int capacity)

- Implement the function: void learnSpell(Spell s)
- Rules: If the spellbook is full, print: SpellBook is full! Otherwise add the spell to the spellbook.

Part 3 - Operator Overloading

Overload + operator for Spell

Two spells can be combined to create a new spell.

Example:

```
Spell s1("FireBlast",50,20,3);
Spell s2("WindSlash",30,10,2);
Spell combo = s1 + s2;
```

Rules for the new spell:

- spellName = "FusionSpell"
- power = sum of powers
- manaCost = sum of manaCosts
- difficulty = average of difficulties

Overload << operator

Allow spells to be printed like this:

```
cout << spell;
```

Output format:

```
Spell: FireBlast | Power:50 | Mana:20 | Difficulty:3
```

Part 4 - Spell Strength Comparison

Overload the > operator.

One spell is considered stronger if its power is greater.

Example:

```
If (spell1 > spell2)
    cout << "Spell 1 is stronger";
```

Part 5 - Magical Stability Rule (The Twist)

Not all spell combinations are safe.

If the combined power exceeds 100, the spell becomes unstable.

In this case the new spell must be automatically renamed to:

"UnstableMagic"

Example:

Spell A power = 60

Spell B power = 50

Combined power = 110

Result spell name = "UnstableMagic"

Students must implement this inside the overloaded + operator.

Part 6 - Display SpellBook

Implement:

```
void displaySpellBook()
```

Example Output:

--- Hermione's SpellBook ---

1. Lumos
2. Expelliarmus
3. FusionSpell

Q2. Image Editor Program

You will build a C++ image editor that reads, manipulates, and saves PGM (Portable Grey Map) image files. The program centers around an Image class that stores pixel data as a 2D integer matrix. Below is a sample pgm image file, and another can be found by clicking here [Sample-2](#):

PGM File Format

A PGM file has this structure:

P2 ← Magic number (always "P2" for ASCII greyscale)

4 4 ← Width (columns) then Height (rows)

255 ← Maximum greyscale value

0 33 0 170 ← Pixel data rows

0 143 0 0

0 0 63 0

170 0 0 255

Each pixel is an integer from 0 (black) to 255 (white).

The Image Class

You must implement a class called Image with the following 8 functions:

Required Functions

Function	Description
1 readImage(filename)	Read a .pgm file into a 2D integer matrix
2 saveImage(filename)	Write the current matrix back to a .pgm file
3 changeBrightness(value)	Add a value (−255 to 255) to every pixel
4 flipImage(direction)	Flip the image horizontally or vertically
5 combineImages(other)	Join two images side by side (same height required)
6 negateImage()	Invert every pixel
7 medianFilter()	Apply a median filter to smooth noise (typically using a 3×3 neighbourhood)
8 applyFilter()	Bonus - implement any additional filter of your choice

Use the following skeleton code as a starting point in order to understand the question more briefly. Code URL: [GitHub](#)